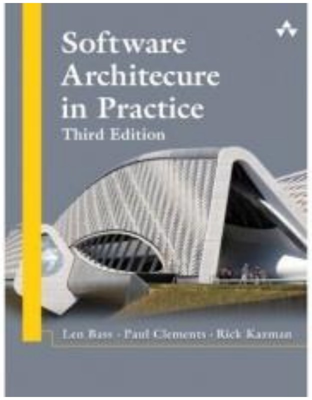
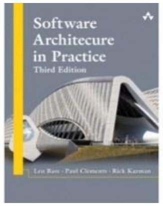


Chapter 5-11: Quality Attributes Scenarios and tactics



Agenda

- Availability
- Performance
- Usability
- Security

Availability Attribute

The ability of the system to mask or repair faults such that the outage period does not exceed a required value over a time period.

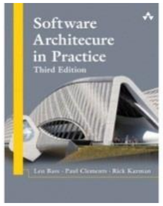
Measure how the system responds to failure.

When the system breaks, how long does it take to resume normal operation?

Stimuli should always be a failure.

Response measures should always include a measure of availability:

availability percentage, time to detect or repair fault, time system in degraded mode, no down time, etc.

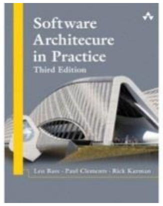


Availability Attributes

The availability of a system can be calculated as the probability that it will provide the specified services within required bounds over a specified time interval.

$$\frac{MTBF}{(MTBF + MTTR)}$$

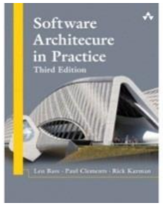
Where: *MTBF* refers to the mean time between failures.
MTTR refers to the mean time to repair.



Availability Attributes

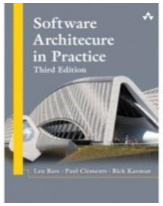
System Availability requirements examples

Availability	Downtime/90 Days	Downtime/Year
99.0%	21 hours, 36 minutes	3 days, 15.6 hours
99.9%	2 hours, 10 minutes	8 hours, 0 minutes, 46 seconds
99.99%	12 minutes, 58 seconds	52 minutes, 34 seconds
99.999%	1 minute, 18 seconds	5 minutes, 15 seconds
99.9999%	8 seconds	32 seconds



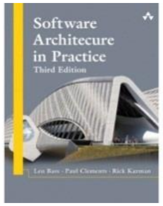
Availability General Scenario

Portion of Scenario	Possible Values
Source	Internal/external: people, hardware, software, physical infrastructure, physical environment
Stimulus	Fault: omission, crash, incorrect timing, incorrect response
Artifact	Processors, communication channels, persistent storage, processes
Environment	Normal operation, startup, shutdown, repair mode, degraded operation, overloaded operation
Response	Prevent the fault from becoming a failure Detect the fault: ▪ Log the fault ▪ Notify appropriate entities (people or systems) Recover from the fault: ▪ Disable source of events causing the fault ▪ Be temporarily unavailable while repair is being effected ▪ Fix or mask the fault/failure or contain the damage it causes ▪ Operate in a degraded mode while repair is being effected
Response Measure	Time or time interval when the system must be available Availability percentage (e.g., 99.999%) Time to detect the fault Time to repair the fault Time or time interval in which system can be in degraded mode Proportion (e.g., 99%) or rate (e.g., up to 100 per second) of a certain class of faults that the system prevents, or handles without failing

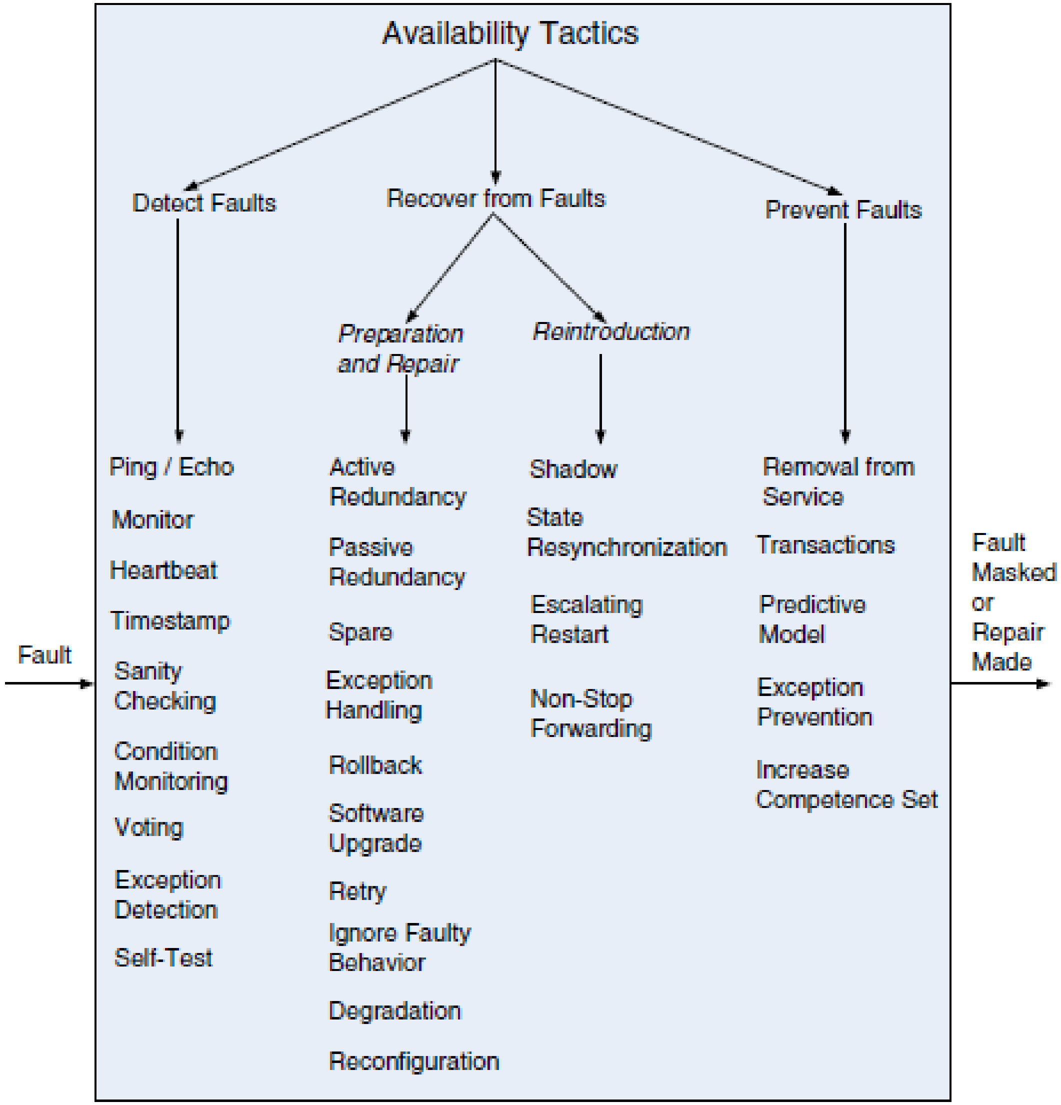


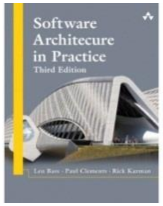
Tactics for availability

- **Availability tactics** may be categorized as addressing one of three categories:
  - Fault detection. Before any system can take action regarding a fault, the presence of the fault must be detected or anticipated
  - Fault recovery. Recover-from-faults tactics are refined into preparation-and-repair tactics and reintroduction tactics
  - Fault prevention. Instead of detecting faults and then trying to recover from them, what if your system could prevent them from occurring in the first place



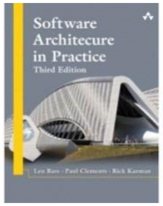
Availability Tactics





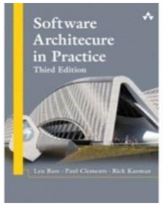
Performance Attributes

- Measure system performance (as opposed to user performance).
- Begins with an event arriving at the system.
  - Responding requires resources (including time) to be consumed.
- Arrival pattern for events can be:
  - Periodic (at regular time intervals)
  - Stochastic (events arrive according to a distribution)
  - Sporadic (unknown timing, but known properties)
    - “No more than 600 per minute”
    - “At least 200 ms between arrival of two events”



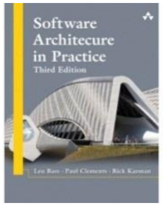
Performance Attributes

- Response measurements include:
  - **Latency:** The time between the arrival of the stimulus and the system’s response to it.
  - **Deadlines in processing:** Points where processing must have reached a particular stage.
  - **Throughput:** Usually number of transactions the system can process in a unit of time.
  - **Response Jitter:** The allowable variation in latency.
  - **Number of events not processed** because the system was too busy to respond.



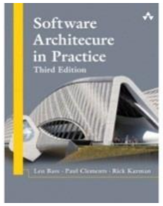
Performance Attributes

- For real-time systems (i.e., embedded devices), measurements are absolute.
  - Look at worst-case scenario.
- For non-real-time systems, measurements should be probabilistic.
  - 95% of the time, the response should be N.
  - 99% of the time, the response should be M.

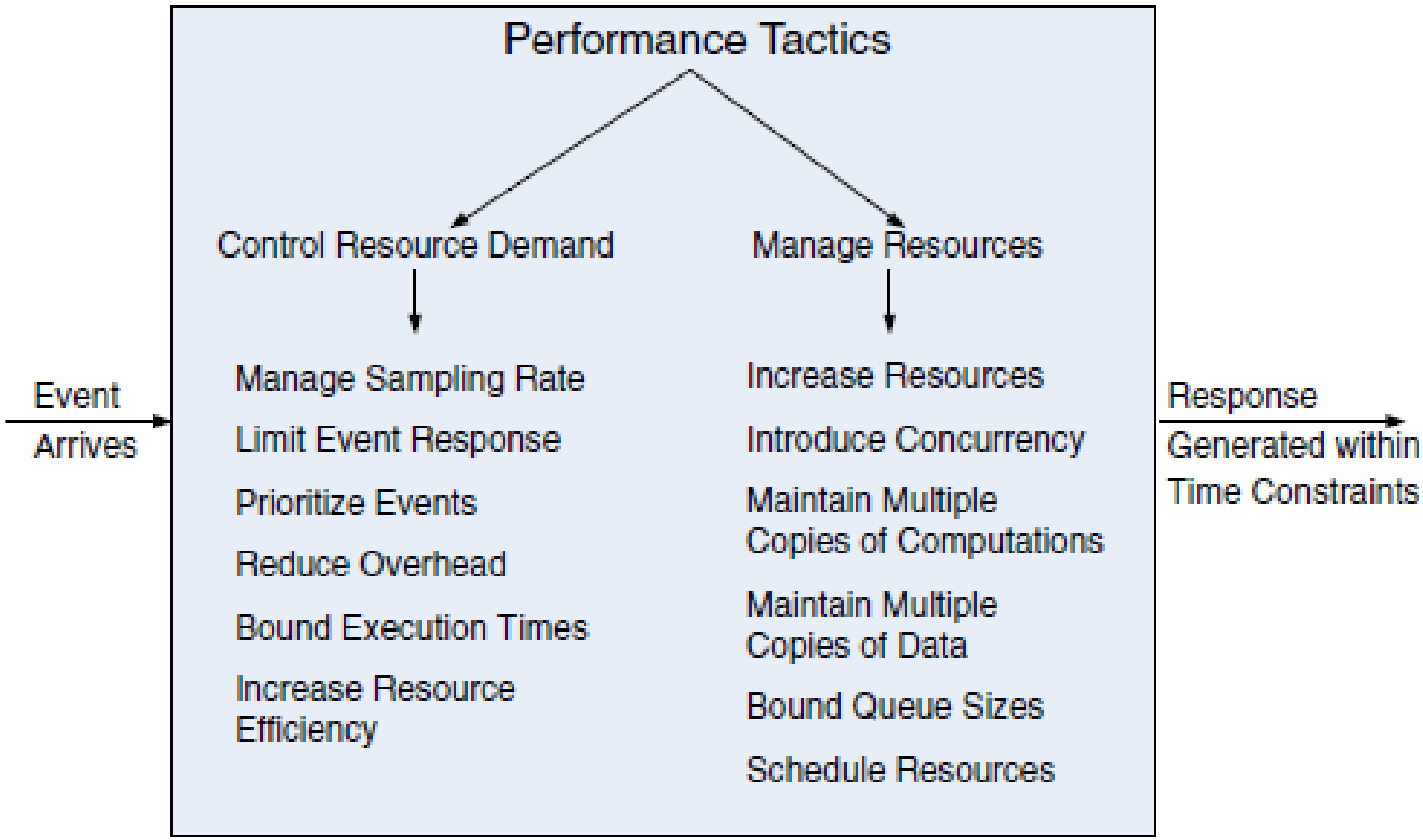


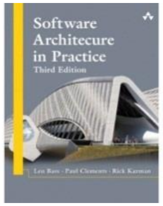
Performance General Scenario

Portion of Scenario	Possible Values
Source	Internal or external to the system
Stimulus	Arrival of a periodic, sporadic, or stochastic event
Artifact	System or one or more components in the system
Environment	Operational mode: normal, emergency, peak load, overload
Response	Process events, change level of service
Response Measure	Latency, deadline, throughput, jitter, miss rate



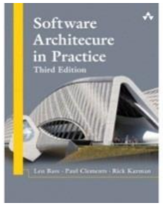
Performance Tactics





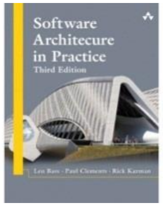
Usability Attributes

- Measure user performance (as opposed to system). Scenarios center around:
  - Learning system features: What can the system do to make learning easier?
  - Using a system effectively: What can the system do to make the user more efficient?
  - Minimizing impact of errors: What can the system do so a user error has minimal impact?
  - Adapting to user needs: Can the system adapt to make a task easier?
  - Increasing confidence and satisfaction: What does the system do to provide feedback?

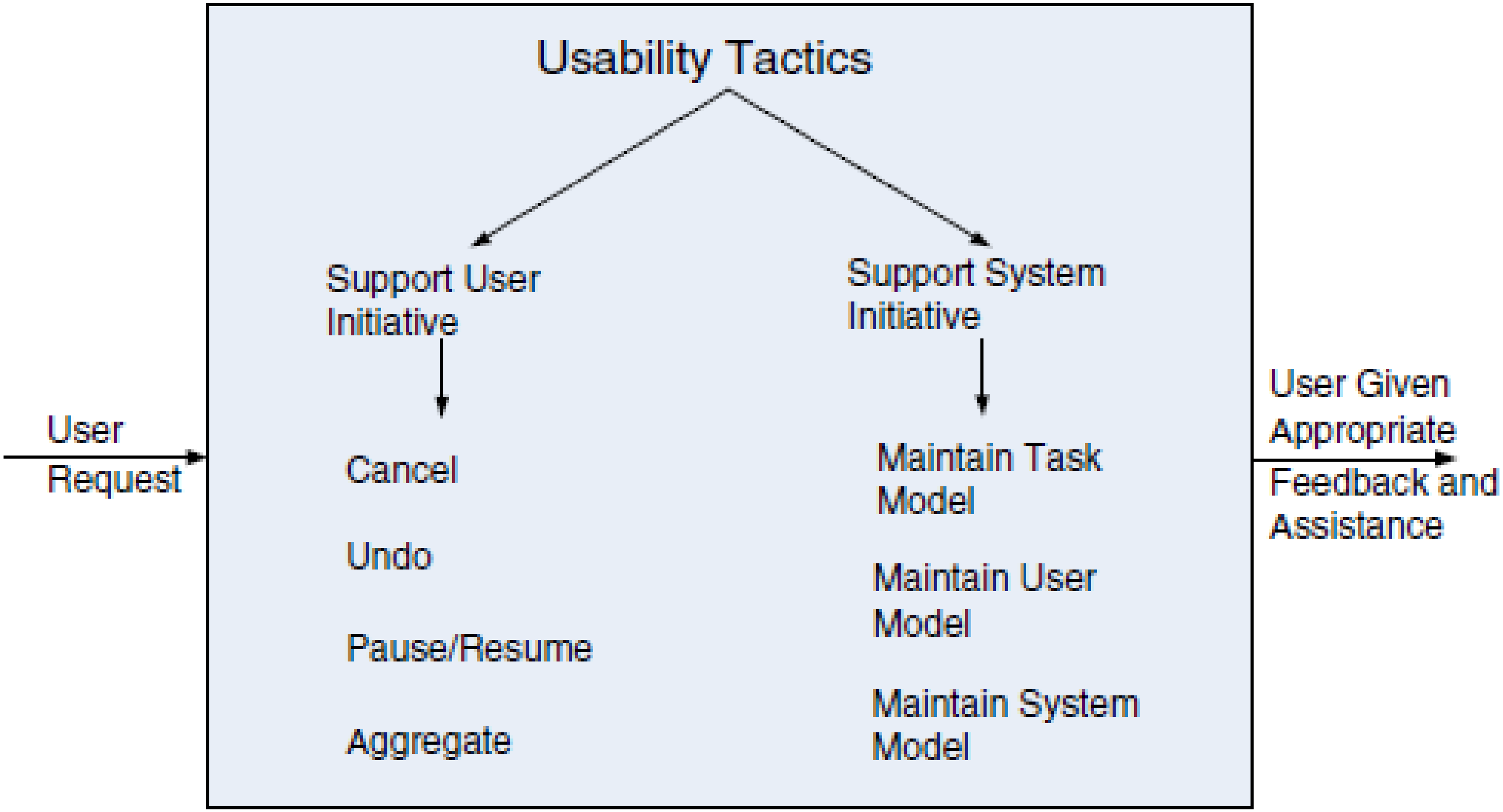


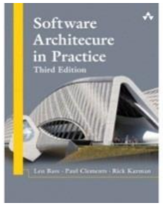
Generic Usability Scenario

Portion of Scenario	Possible Values
Source	End user, possibly in a specialized role
Stimulus	End user tries to use a system efficiently, learn to use the system, minimize the impact of errors, adapt the system, or configure the system.
Environment	Runtime or configuration time
Artifacts	System or the specific portion of the system with which the user is interacting
Response	The system should either provide the user with the features needed or anticipate the user's needs.
Response Measure	One or more of the following: task time, number of errors, number of tasks accomplished, user satisfaction, gain of user knowledge, ratio of successful operations to total operations, or amount of time or data lost when an error occurs



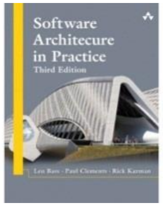
Usability Tactics





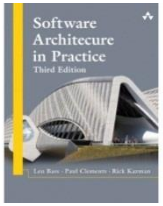
Security Quality Scenarios

- Measure of the system’s ability to protect data from unauthorized access while still providing service to authorized users.
- Scenarios measure response to attack.
  - Stimuli are attacks from external systems/users or demonstrations of policies (log-in, authorization).
- Responses include auditing, logging, reporting, analyzing.
  - Response measures include amount of data loss/compromise, time to detect/mitigate, % of attacks resisted, etc.



Security General Scenario

Portion of Scenario	Possible Values
Source	Human or another system which may have been previously identified (either correctly or incorrectly) or may be currently unknown. A human attacker may be from outside the organization or from inside the organization.
Stimulus	Unauthorized attempt is made to display data, change or delete data, access system services, change the system's behavior, or reduce availability.
Artifact	System services, data within the system, a component or resources of the system, data produced or consumed by the system
Environment	The system is either online or offline; either connected to or disconnected from a network; either behind a firewall or open to a network; fully operational, partially operational, or not operational.
Response	Transactions are carried out in a fashion such that ▪ Data or services are protected from unauthorized access. ▪ Data or services are not being manipulated without authorization. ▪ Parties to a transaction are identified with assurance. ▪ The parties to the transaction cannot repudiate their involvements. ▪ The data, resources, and system services will be available for legitimate use. The system tracks activities within it by ▪ Recording access or modification ▪ Recording attempts to access data, resources, or services ▪ Notifying appropriate entities (people or systems) when an apparent attack is occurring
Response Measure	One or more of the following: ▪ How much of a system is compromised when a particular component or data value is compromised ▪ How much time passed before an attack was detected ▪ How many attacks were resisted ▪ How long does it take to recover from a successful attack ▪ How much data is vulnerable to a particular attack



Security tactics